National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Multivariable Calculus	Course Code:	MT2008
	Program:	Electrical Engineering	Semester:	Fall 2023
	Duration:	3 hour	Total Marks:	70
	Exam Date:	December 30th, 2023	Weight:	50
	Section:	All	Page(s):	
	Exam Type:	Final	CLO #	1, 2, 3

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Instruction/Notes: 1. Do not forget to write your Name and Roll Numbers.

Question 1: (CLO 01) (marks: 5+5+5)

a) Formulate the parametric equations and symmetric equations for the line from $(1,-1,1)$ and parallel to the line $x+2=\frac{1}{2}y=z-3$.

b) Find an equation of the plane that passes through the point $(1,-1,1)$ and contains the line with symmetric equations $x=2y=3z$.

c) Calculate the distance between the given parallel planes $6z=4y-2x$, $9z=1-3x+6y$.

Question 2: (CLO 02) (marks: 5+10+10)

a) If the resistors of R_1, R_2, and R_3 ohms are connected in parallel to make an R-ohm resistor, the value of R can be found from the equation

$$\frac{1}{R}=\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_3}$$

Find the value of $\partial R/\partial R_2$ when $R_1=30$, $R_2=45$, and $R_3=90$ ohms.

b) Suppose that the temperature at a point (x,y,z) in space is given by $T(x,y,z)=200e^{-x^2-3y^2-9z^2}$ where T is measured in degrees Celsius and x,y,z in meters. In which direction does the temperature increase fastest at the point $(1,1,-2)$? What is the maximum rate of increase?

c) A delivery company accepts only rectangular boxes the sum of whose length and girth (perimeter of a cross-section) does not exceed 108 inches. Find the dimensions of an acceptable box of largest volume.

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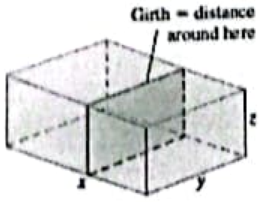

Question 3: (CLO 03) (marks: 10+10+10)

a) Find area of the part of the surface $z = xy$ that lies within the cylinder $x^2 + y^2 = 1$.

b) Use Green's Theorem to calculate the work done by forçe field in moving a particle over a triangle from (0 , 0) to (0 , 4) to (2 , 0) to (0 , 0) where $F = (y\ cosx - xy\ sinx,\ xy + x\ cosx)$. (Check the orientation of the curve before applying the theorem.)

c) Calculate the flux of vector field $F(x, y, z) = 3xy^2\mathbf{i} + xe^z\mathbf{j} + z^3\mathbf{k}$ across the surface of the solid bounded by the cylinder $y^2 + z^2 = 1$ and planes $x = -1$ and $x = 2$.